

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 18
 - Connectedness
 - Differentiation

Warm-up

Let (S, d) be a metric space. Prove that if $f : S \rightarrow \mathbb{R}$ and $g : S \rightarrow \mathbb{R}$ are continuous at $a \in S$, then so is $f + g$.

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Let (S, d) be a metric space. Prove that if $f : S \rightarrow \mathbb{R}$ and $g : S \rightarrow \mathbb{R}$ are continuous at $a \in S$ with $g(a) \neq 0$, then so is f/g .

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Connected spaces

Definition

A metric space (S, d) is called **disconnected** if there exists two nonempty, disjoint open subsets $U, V \subseteq S$ with $S = U \cup V$.

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Examples:

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Examples:

- the real line with the Euclidean metric $(\mathbb{R}, d_{eucl})$ is connected
- the real line with the discrete metric $(\mathbb{R}, d_{disc})$ is connected
- euclidean space $\mathbb{R}^n$ is connected
- an interval $I \subseteq \mathbb{R}$ with the Euclidean metric is connected

Challenge!

Problem

Prove that the set $\mathbb{Q} \subseteq \mathbb{R}$ is disconnected.

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

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Suppose that $\mathbb{R}$ is disconnected.

Then $\mathbb{R} = U \cup V$ where:

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Then $\mathbb{R} = U \cup V$ where:

- $U \neq \emptyset, V \neq \emptyset$

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Suppose that $\mathbb{R}$ is disconnected.

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- $U \cap V = \emptyset$

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- $U \cup V = \mathbb{R}$

In particular, U is a nonempty, proper subset of $\mathbb{R}$ which is both open and closed. □

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By the Representation Theorem for Open Sets:

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$$U = (a_1, b_1) \cup (a_2, b_2) \cup \dots$$

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By the Representation Theorem for Open Sets:

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is a union of disjoint open intervals.

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In particular, b_1 is an accumulation point of U but not an element of U .

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In particular, b_1 is an accumulation point of U but not an element of U .

This contradicts the fact that U is closed. □

Images of connected sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \rightarrow T$ be a function.

If f is continuous and S is connected, then $f(S)$ is connected.

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Suppose $f(S)$ is disconnected.

Then $f(S) = U \cup V$ for nonempty, disjoint, open sets
 $U, V \subseteq f(S)$.

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Suppose $f(S)$ is disconnected.

Then $f(S) = U \cup V$ for nonempty, disjoint, open sets
 $U, V \subseteq f(S)$.

Then $f^{-1}(U)$ and $f^{-1}(V)$ are non-empty, disjoint, open sets and

$$f^{-1}(U) \cup f^{-1}(V) = f^{-1}(U \cup V) = f^{-1}(f(S)) = S.$$

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This is a contradiction. □

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We can make any interval as a continuous image:

$$f : \mathbb{R} \rightarrow \mathbb{R}$$
$$f(x) = \frac{b+a}{2} + \frac{(b-a)x}{1+x^2}$$

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We can make any interval as a continuous image:

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = a + \frac{2(b-a)x^2}{1+x^4}$$

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Intermediate value theorem

Theorem (Intermediate Value Theorem)

*Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function.
Then for any value y between $f(a)$ and $f(b)$, there exists
 $c \in [a, b]$ with $f(c) = y$.*

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Let y be between $f(a)$ and $f(b)$, and suppose that $y \notin J$.

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Let y be between $f(a)$ and $f(b)$, and suppose that $y \notin J$.

Without loss of generality, assume $f(a) < f(b)$.

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Let $J = f([a, b])$.

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Let y be between $f(a)$ and $f(b)$, and suppose that $y \notin J$.

Without loss of generality, assume $f(a) < f(b)$.

Then $J = U \cup V$ for

$$U = (-\infty, y) \cap J \quad \text{and} \quad V = (y, \infty) \cap J$$

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These are nonempty, disjoint, and open in the subspace J of $\mathbb{R}$.

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This is a contradiction! □

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- Differentiation

Differentiability

Definition

A real-valued function $f(x)$ defined on an open interval (a, b) is **differentiable** at $c \in (a, b)$ if

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \text{ exists.}$$

In this case the limit is called the derivative of f at c , and denote by $f'(c)$.

Chord function

Theorem (Helper Theorem)

Let f be a real-valued function on (a, b) . Then f is differentiable at $c \in (a, b)$ if and only if there exists a function f^ on (a, b) which is continuous at c and satisfies*

$$f(x) - f(c) = (x - c)f^*(x).$$

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The function $f^*(x)$ is the **chord slope function**.

Chord function

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Suppose $f(x)$ is differentiable at c .

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Suppose $f(x)$ is differentiable at c . Define

$$f^*(x) = \begin{cases} \frac{f(x)-f(c)}{x-c}, & x \neq c \\ f'(c), & x = c \end{cases}$$

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Then $\lim_{x \rightarrow c} f^*(x) = f'(c)$, so f^* is continuous at c .

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Conversely, suppose f^* exists and is continuous at c .

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Then $\lim_{x \rightarrow c} f^*(x) = f'(c)$, so f^* is continuous at c .

Conversely, suppose f^* exists and is continuous at c .

Then

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c} f^*(x) = f^*(c).$$

In particular f is differentiable at c and $f'(x) = f^*(c)$. □

Differentiable implies continuous

Theorem

*Suppose $f(x)$ is a real valued function on (a, b) and is differentiable at $c \in (a, b)$.
Then f is continuous at c .*

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Proof.

By the Helper Theorem, there exists a function f^* on (a, b) which is continuous at c with

$$f(x) = (x - c)f^*(x).$$

Differentiable implies continuous

Theorem

*Suppose $f(x)$ is a real valued function on (a, b) and is differentiable at $c \in (a, b)$.
Then f is continuous at c .*

Proof.

By the Helper Theorem, there exists a function f^* on (a, b) which is continuous at c with

$$f(x) = (x - c)f^*(x).$$

Since $(x - c)$ and f^* are both continuous at c , so is $f(x)$. □

Additivity of derivatives

Theorem

Let f, g be real functions on (a, b) which are differentiable at $c \in (a, b)$.

$$(f \pm g)'(c) = f'(c) \pm g'(c)$$

Additivity of derivatives

Proof.

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Proof.

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Additivity of derivatives

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$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Therefore

$$(f \pm g)(x) - (f \pm g)(c) = (x - c)(f^*(x) \pm g^*(x)).$$

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Since f^* and g^* are continuous at c , so is $f^* \pm g^*$.

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Therefore

$$(f \pm g)(x) - (f \pm g)(c) = (x - c)(f^*(x) \pm g^*(x)).$$

Since f^* and g^* are continuous at c , so is $f^* \pm g^*$.

The Helper Theorem implies that $g \pm f$ is differentiable and

$$(f \pm g)'(c) = f'(c) \pm g'(c)$$



Product rule

Theorem

Let f, g be real functions on (a, b) which are differentiable at $c \in (a, b)$.

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c)$$

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$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Therefore

$$\begin{aligned}(fg)(x) - (fg)(c) &= f(x)g(x) - f(c)g(c) \\&= f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c) \\&= f(x)(g(x) - g(c)) + (f(x) - f(c))g(c) \\&= (x - c)[f(x)g^*(x) + f^*(x)g(c)]\end{aligned}$$

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Helper Theorem says:

$$(fg)'(c) = f(c)g^*(c) + f^*(c)g(c) = f(c)g'(c) + f'(c)g(c).$$

Quotient rule

Theorem

Let f, g be real functions on (a, b) which are differentiable at $c \in (a, b)$ with $g(c) \neq 0$.

$$(f/g)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}$$

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Quotient rule

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$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

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Helper Theorem says:

$$(f/g)'(c) = \frac{f^*(c)g(c) - f(c)g^*(c)}{g(c)^2} = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}.$$

Chain rule

Theorem

Let f be a real function on an interval $I = (a, b)$, and let g be a real function on $f(I)$, and consider the composition

$$h = g \circ f : (a, b) \rightarrow \mathbb{R}.$$

Suppose $c \in (a, b)$ and $f(c)$ is an interior point of $f(I)$. If f is differentiable at c and g is differentiable at $f(c)$, then h is differentiable at c and

$$h'(c) = (g \circ f)'(c) = g'(f(c))f'(c).$$